LINGVESSA

Reference Module Design

English A1 M01 + Čeština A1 M01 / v1.6

Implementation-ready design of the two launch reference modules: lesson sequences, two-level mastery evidence, answer evaluation, pronunciation UX, optional grounded AI extensions, review, media, accessibility and regression tests.

REFERENCE MODULES E-A1-M01 / C-A1-M01

SHARED ENGINE LESSON + REVIEW + MASTERY + MEDIA + OPTIONAL AI

GOVERNING BASELINE FORMAL v1.7 / TECHNICAL v1.7 / TECH DESIGN v1.7 / CURRICULUM A1 v1.6 / AI v1.4 / PLAN v1.4

PRONUNCIATION UX USABLE SPEECH: CORRECT / TRY AGAIN; FAILURE: SPEECH_UNAVAILABLE

CORE GUARANTEE COMPLETE LEARNING PATH WITHOUT AI OR PAID PROVIDER

Status: OWNER-LOCKED IMPLEMENTATION BASELINE
Updated: 9 September 2026 | Replaces Reference Module Design v1.5

Паспорт документа

Поле	Значення
Ідентифікатор	LINGVESSA-REF-A1-M01
Версія / дата	1.6 / 09.09.2026
Статус	OWNER-LOCKED IMPLEMENTATION BASELINE; owner lock 10.09.2026; P0 authorized.
Замінює	Reference Module Design v1.5; previous versions remain historical.
Governing documents	Formal Specification v1.7; Technical Specification v1.7; Technical Design v1.7; Curriculum Blueprint A1 v1.6; Reference Module Design v1.6; AI Integration Specification v1.4; Implementation Plan v1.4 (execution order only).
Reference units	English E-A1-M01 Hello & first contact; Čeština C-A1-M01 Dobrý den!
Склад	4 lessons + 1 Module Recap на кожну мову; knowledge inventory; review seed; media manifest; optional AI extensions; regression suite.
Межа твердження	Документ є design fixture. Він не доводить, що ще не реалізований код уже пройшов QA.

Що змінено у v1.6

Зміна	Результат
Product identity	Увесь документ синхронізовано з робочою назвою Lingvessa.
Dependencies	Baseline оновлено до Formal v1.7, Technical v1.7, Technical Design v1.7, Curriculum A1 v1.6 та AI Integration v1.4.
Speech UX	Learner-facing binary feedback повернуто для usable speech: correct / try again. Low confidence, mic denial, timeout і provider failure = speech_unavailable; жодного hard gate чи punitive mastery.
Mic behavior	Після явного натискання microphone control переходить у recording/pulsing state; reduced-motion має textual/static alternative.
Mastery	Шість глобальних skill projections відокремлено від knowledge-item facets: meaning, recognition, recall, spelling, production, listening recognition, pronunciation target feature, context use, register тощо.
Result UX	Lesson result більше не зобов’язаний показувати всі шість skills. Він показує mistakes, completion і релевантний progress; повний profile доступний у Progress.
AI learning	Додано optional grounded extension points: explain differently, adaptive hints, extra practice, writing feedback, bounded role-play. AI не є dependency core lesson.
AI administration/media	Generation/QA може використовувати AI через provider-agnostic pipeline, але published content/assets проходять ті самі approval gates.
Evidence safety	Free-form AI practice не змінює mastery без approved evaluator/rubric; AI failure/quota exhaustion не створює negative evidence.
Regression	Suite розширено до 32 scenarios, додано pronunciation feedback states, two-level mastery, AI fallback/grounding/evidence isolation та result UX.

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ЧАСТИНА A. КОНТРАКТ REFERENCE MODULES

1. Призначення, межі та статус

Цей документ деталізує перші модулі A1 до рівня, достатнього для прямої production-oriented reference implementation. Його мета - перевірити на реальному контенті спільний engine contract для English і Čeština, а не оголосити архітектуру перевіреною до появи коду, tests і QA evidence.

Architecture invariant
English і Čeština проходять через один Lesson Engine, Review Engine, two-level Mastery model, Media Pipeline та спільний optional AI Layer. Language-specific behavior задається content configuration, evaluator rules, locale/pronunciation strategies та grounded knowledge. Окремий Czech engine fork заборонений.

- English reference module: E-A1-M01 Hello & first contact.
- Czech reference module: C-A1-M01 Dobrý den!.
- Кожний модуль містить 4 core lessons + 1 integrated Module Recap.
- Усі dialogues, examples, activities, characters, images and audio scripts are original or properly licensed; no Babbel copying.
- Required media is pre-generated or programmatic, QA-approved and bound by asset_id + asset_version before publish.
- AI extensions are optional. The deterministic non-AI path remains complete and is the acceptance baseline.
- Після implementation QA ці модулі залишаються regression fixtures для schema, evaluator, mastery, AI adapter and responsive UI changes.

Не робимо на цьому етапі
Не створюємо весь A1, не оптимізуємо production hosting, не фіксуємо Level Test thresholds і не робимо paid AI mandatory. Reference implementation має довести correctness фундаменту до масштабування M02-M12.

2. Acceptance criteria and engine coverage

Capability	English proof	Czech proof	Required implementation evidence
Content hierarchy	E-A1-M01	C-A1-M01	course -> level -> module -> lesson -> activity -> item; all nodes versioned
Vocabulary / chunks	contractions, repair	register, fixed case chunks	knowledge_item IDs, prerequisites, active/receptive scope
Interactive grammar	be, a/an, questions	být, ty/vy, origin patterns	staged explanation, language-aware accepted variants
Image / matching	greetings and profiles	register and profiles	original/programmatic assets with alt/decorative metadata
Typing / builder	apostrophe and spelling	diacritics and morphology	Unicode-aware normalization; objective-specific strictness
Listening/listening_recognition	gist/detail and spelling	gist/detail and vowel/stress awareness	normal + slow audio; transcript policy; multiple voices
Dialogue chain	neutral first contact	formal/informal branches	persistent prior turns; speaker/voice/avatar synchronization
Speech practice	greetings and intro	stress, length, salient forms	mic pulsing; usable correct/try again; unavailable neutral; retry/continue
Writing / online	sample profile	sample profile	no personal data required; explicit Writing evidence
Remediation	error categories	form/register categories	correct-or-reveal for scored tasks; immutable lapse history
Mastery	6 global skills + item facets	same shared contract	evidence target = skill + relevant facet; no cross-language state
Review seed	facet-aware due/weak/error	facet-aware due/weak/error	scheduling within same target language
AI extension	grounded micro-explain / extra practice	grounded register/diacritics explain	optional provider-neutral capability + scripted/non-AI fallback
Accessibility	same semantics	same semantics	keyboard/touch, transcript/caption, non-mic path, reduced motion

Capability	English proof	Czech proof	Required implementation evidence
Media pipeline	en-GB/en-US metadata	cs-CZ metadata	asset_id + asset_version, provenance, rights, QA and publish gate

2.1. Normative requirements

ID	Вимога	Пріоритет	Критерій перевірки
REF-A1-001	Both modules use one versioned content schema and shared renderers.	MUST	Schema validation passes without language-specific engine fork.
REF-A1-002	Progress, mastery and review are isolated by language_profile/user_course and target language.	MUST	English/Czech switching preserves independent state.
REF-A1-003	Every lesson has context, active retrieval and visible result/remediation.	MUST	Manifest validator finds all required stages.
REF-A1-004	Scored items require correct or reveal before completion.	MUST	Next/auto-advance state tests pass.
REF-A1-005	Reveal and skip do not count as independent mastery.	MUST	Evidence and scheduler events contain reason/rule version.
REF-A1-006	Usable speech may show correct/try again; unavailable speech is neutral and non-blocking.	MUST	Correct, mismatch, denial, low-confidence and provider-failure paths complete safely.
REF-A1-007	Mastery uses 6 global skill projections plus item-level facets.	MUST	Meaning/spelling/listening/pronunciation/register evidence remains distinguishable.
REF-A1-008	English variants are en-GB first and en-US aware.	MUST	Selected variety controls voices, notes and accepted variants.
REF-A1-009	Czech evaluation preserves diacritics, morphology and register distinctions.	MUST	Strict/form-focused and meaning-focused tests produce different facet evidence.
REF-A1-010	Required assets are resolved and approved before publish.	MUST	asset_id + asset_version + QA + provenance/rights all resolve.
REF-A1-011	No learner flow requires the user to source media or provide PII.	MUST	Sample profiles and system media cover every required path.
REF-A1-012	Reference modules remain regression fixtures after approval.	MUST	Schema/evaluator/UI/AI-adapter changes run the regression suite.
REF-A1-013	AI is optional, grounded and never a prerequisite for core completion.	MUST	AI disabled/outage/quota tests preserve complete non-AI lesson path.
REF-A1-014	Free-form AI output cannot directly mutate mastery or published content.	MUST	Only approved evaluator/rubric/workflow transitions can create bounded evidence/publish change.

3. Shared content and runtime rules

Rule	Specification
Lesson sequence	context -> input/noticing -> concise Ukrainian explanation -> controlled practice -> active retrieval -> listening/communication -> result/remediation
One primary action	One screen has one main cognitive action. Long explanations use staged reveal or short linked cards.
Hints	Support decreases from introduction to recognition, guided retrieval and independent typing/speaking.

Rule	Specification
Next	Scored: unavailable until correct or revealed. Speech: allowed after usable feedback or speech_unavailable according to activity policy.
Incorrect	Objective-specific feedback; error category; retry/reveal. Speech mismatch may show try again without becoming a hard gate.
Auto-advance	Only from an allowed completion state; never converts error, timeout or skip into success.
Audio	Relevant words/sentences/dialogues provide normal and slow playback with transcript metadata.
Translation	UI/explanations use uk-UA; target-language objects never depend on hardcoded Ukrainian fields.
Personalisation	Name/country inserted only with consent; complete fictional sample profile always exists.
Media	Runtime consumes approved assets. Live generation is not a dependency of a normal lesson attempt.
AI	Optional capability is invoked only through approved AI Gateway/capability policy; output is grounded, bounded and has non-AI fallback.
Versioning	Attempt, evaluator, content, asset, mastery, scheduler and AI capability/model metadata retain concrete versions where relevant.

4. Knowledge, evidence and ID contract

Object	Required fields / rule
knowledge_item	stable ID, language, type, canonical form/meaning, active/receptive scope, prerequisites, module/lesson links
accepted_variant_set	objective, language/variety/register, normalization policy, permitted forms, evaluator version
activity_item	objective, operation, prompt, answer/evaluator, support level, required assets, evidence mapping
attempt	immutable item/content/evaluator versions, response metadata/hash, outcome, support/reveal/skip flags, error category, timestamps
evidence_event	global_skill, facet, knowledge_item, quality/independence, rule/model version, source attempt; no cross-language target
mastery_state	profile + knowledge_item + facet estimate/uncertainty plus derived global skill aggregation; versioned model
review_candidate	user_course/language, knowledge item + facet, due reason, source evidence, scheduler version
media_binding	concrete asset_id + asset_version; role/turn binding; accessibility and rights status
ai_interaction	capability, provider/model, grounding refs, prompt_template version, output safety/validation state, cost/quota metadata; no raw secret

Group notation
Ranges such as CZE_K_NUM_00..20 or wildcard-like media patterns are authoring shorthand only. Build MUST expand them to concrete atomic IDs and reject unresolved ranges/wildcards before publish.

4.1. Two-level mastery and facet evidence

Lingvessa separates learner-facing/aggregate skills from the specific capability proven for an individual knowledge item. A global skill projection is derived from many item/facet evidence events; it is never a substitute for them.

Layer	Values / examples	Rule
Global skill projections	Vocabulary, Grammar, Listening, Speaking, Writing, Pronunciation	Used for Progress/readiness summaries and adaptive balance; derived, not primary evidence store.

Layer	Values / examples	Rule
Lexical facets	meaning, recognition, recall_l2_to_l1, recall_l1_to_l2, spelling, production, listening_recognition, context_use, register	Only relevant facets exist for a knowledge item; one correct card does not master all.
Grammar/morphology facets	grammar_form, meaning, context_use, morphology, word_order, register	Evidence target follows objective; Czech form error does not erase lexical meaning.
Pronunciation facets	pronunciation_target, intelligibility	Usable speech can add quality-tagged evidence; low confidence/provider failure adds none.
Interaction facets	interaction_repair, interaction_goal, register, context_use	May aggregate into Speaking/Grammar/Vocabulary depending on evaluator map.

4.2. Optional AI extension contract

Capability	Allowed in reference modules	Evidence / fallback rule
Explain differently	Explain approved grammar/usage/diacritics in Ukrainian with simple examples.	Must cite internal grounding refs; viewing is no mastery; fallback = authored explanation.
Adaptive hint	Rephrase a hint or select next scaffold within declared hint_policy.	Marks assistance level; cannot expose answer before policy; fallback = authored hint.
Extra practice	Generate bounded practice from approved knowledge items and weak facet.	Free-form = practice-only; validated deterministic template may create bounded evidence.
Writing feedback	Comment on short fictional-profile or self-introduction text.	Advisory by default; cannot overwrite deterministic answer history; fallback = rubric/sample.
AI role-play	Short first-contact scenario after core scripted dialogue.	Optional, bounded turns/lexis; scripted role-play fallback; no core progression dependency.
Admin QA/generation	Draft assets/content checks behind workflow.	Never direct publish; all output versioned/provenance-reviewed.

ЧАСТИНА B. ENGLISH REFERENCE MODULE

5. English reference module E-A1-M01

Field	Value
Title / scenario	Hello & first contact / first day at an international language club
Can-do	Greet, introduce yourself, ask/give name and origin, use repair phrases and close a short first-contact exchange.
Variant	en-GB primary; en-US selectable; meaningful differences use metadata, accepted variants or concise notes.
Grammar	be with I/you/he/she; subject pronouns; a/an; basic be questions and short answers; contractions.
Lexis	greetings, names, selected countries/languages, roles and classroom-survival/repair phrases.
Pronunciation	alphabet/letter names as support; /h/ awareness; contractions; clear sentence stress; intelligibility focus.
AI extensions	optional explanation/hints/extra practice/bounded first-contact role-play grounded in module knowledge.
Units	E-A1-M01-L01..L04 + E-A1-M01-R01 Module Recap.

5.1. English knowledge inventory

stable_code / group	Type	Canonical form	Scope	Function / evidence
ENG_K_GREET_HELLO	chunk	Hello.	active	neutral greeting; Vocabulary/meaning + production
ENG_K_GREET_HI	chunk	Hi.	active	informal greeting; meaning/register
ENG_K_GREET_MORNING	chunk	Good morning.	active	time-bound greeting
ENG_K_GREET_EVENING	chunk	Good evening.	active	time-bound greeting
ENG_K_BYE	variant set	Goodbye. / Bye.	active	leave-taking; register note
ENG_K_THANKS / PLEASE / SORRY	3 chunks	Thank you. / please / Sorry.	active	politeness and repair
ENG_K_REPEAT	chunk	Can you repeat, please?	active	repair / clarification
ENG_K_DONT_UNDERSTAND	chunk	I don't understand.	active	repair strategy
ENG_K_NAME_Q	chunk	What's your name?	active	personal information
ENG_K_NAME_A	variant set	My name's... / I'm...	active	personal information
ENG_K_NICE_MEET	chunk	Nice to meet you.	active	social close
ENG_K_NICE_MEET_TOO	chunk	Nice to meet you too.	active	response
ENG_K_FROM_Q / FROM_A	2 chunks	Where are you from? / I'm from...	active	origin exchange
ENG_K_SPELL_Q	chunk	How do you spell it?	support	clarification and spelling
ENG_K_LANG_A	pattern	I speak [language].	receptive-active	language information
ENG_K_BE_I / YOU / HE / SHE	4 grammar	I am / you are / he is / she is	active	Grammar/ grammar_form + contractions
ENG_K_BE_Q	grammar	Are you...? / Is he/she...?	active	controlled questions with be
ENG_K_YESNO	variant set	Yes, I am. / No, I'm not.	active	short answers
ENG_K_A_AN	grammar	a / an	active	role nouns; sound-based choice
ENG_K_PRONOUNS	4 items	I / you / he / she	active	speaker/character perspective

stable_code / group	Type	Canonical form	Scope	Function / evidence
ENG_K_ROLE_STUDENT / TEACHER / ENGINEER	3 lexical	student / teacher / engineer	active	identity roles + a/an
ENG_K_COUNTRY_GB / UA / CZ / PL / DE / ES	6 lexical	the UK / Ukraine / Czechia / Poland / Germany / Spain	active/support	origin values; atomic IDs
ENG_K_LANG_EN / UK / CS / PL / DE / ES	6 lexical	English / Ukrainian / Czech / Polish / German / Spanish	receptive-active	language values
ENG_K_CITY_LON / PRA / KYI / WAR	4 lexical	London / Prague / Kyiv / Warsaw	support	spelling/profile values
ENG_K_PRON_H	pronunciation	/h/ in hello / hi	awareness	perception + pronunciation_target practice
ENG_K_PRON_CONTRACTION	pronunciation	I'm / you're / he's / she's	awareness	connected speech
ENG_K_PRON_STRESS	pronunciation	short introduction rhythm	awareness	intelligibility
ENG_K_ORTH_APOSTROPHE	orthography	straight/curly apostrophe normalization	active	spelling/form task only when objective requires
ENG_K_ORTH_SPELLING	orthography	letter names and short names/cities	support	listening/spelling evidence

6. English Lesson 1 - Hello!

Objective: recognise and use basic greetings/leave-taking plus two repair phrases; notice /h/ in frequent greetings.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Context / scene	View an original club-arrival scene and hear two short turns.	No answer; Hello / Hi	intro -> completed	IMG_ENG_M01_CLUB_01 v1; dialogue; no mastery
2	Vocabulary intro	Listen to Hello, Hi, Good morning/evening, Goodbye and Bye.	tap audio; uk-UA gloss optional	each card completes after exposure	6 audio assets; recognition seed
3	Image choice	Choose greeting for morning/evening scene.	Good morning / Good evening	incorrect -> cue -> retry/reveal	2 generated scenes; Vocabulary/meaning
4	Listening match	Hear greeting without text and choose card.	Hello / Hi / Goodbye / Bye	retry; deterministic reveal	normal + slow; Listening/listening_recognition
5	Usage note	Read concise neutral/informal note.	No answer	staged note completed	text alternative; no mastery
6	Word builder	Build HELLO after audio.	hello	case-insensitive; correct-or-reveal	Writing/spelling + audio
7	Pronunciation practice	Listen and repeat Hello / Hi; press mic; control pulses while recording.	model phrase; /h/ focus	usable -> correct / try again; unavailable -> neutral; retry/continue	reference audio; Pronunciation/pronunciation_target only if usable
8	Survival intro	Meet Thank you, Sorry, I don't understand, Can you repeat, please?	exposure + audio	cards complete after exposure	4 audio assets; chunk recognition
9	Classification	Sort thanks / misunderstanding / repeat-request phrases.	3 functional categories	resolve all or reveal	programmatic icons; context_use

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
10	Dialogue chain	Choose a repair phrase after a teacher speaks too quickly.	Can you repeat, please? / I don't understand.	wrong choice keeps turn open; correct/reveal continues	2 voices + avatars; Vocabulary/context_use (selection tests repair-phrase understanding; no Speaking evidence)
11	Retrieval check	Translate Thank you / Goodbye / Can you repeat, please? with decreasing support.	configured target variants	guided first; final typed; reveal tracked	Vocabulary/ recall_l1_to_l2 + Writing/production
12	Lesson result	Review scored mistakes and start remediation if needed.	0 unresolved scored items	show mistakes/completion + relevant progress; full skill profile in Progress	lesson_completed + immutable attempts

7. English Lesson 2 - I am / You are

Objective: build simple be sentences for I/you/he/she, recognise contractions and use a/an with basic roles.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Character context	See four recurring profiles and hear short introductions.	I'm Maya; You're Daniel; He's a teacher; She's a student.	exposure completed	4 avatars + audio; no mastery
2	Interactive grammar	Complete I ___ Maya / You ___ Daniel.	am / are	correct reveals concise pattern note	Grammar/ grammar_form
3	Full-to-contraction match	Match I am/you are/he is/she is to contractions.	4 matches	resolve all; apostrophe clear	audio examples; Grammar/grammar_form
4	Pronoun match	Map I/you/he/she to speaker/addressee/ character.	context-dependent	wrong highlights perspective	avatars; Vocabulary/meaning + Grammar
5	a/an discovery	Compare a student, a teacher, an engineer and choose article.	a / an	brief sound-based rule after response	3 original role images; Grammar/context_use
6	Gap fill	Complete She ___ a student / He ___ an engineer.	is / is	retry; correct-or-reveal	Grammar/ grammar_form
7	Contraction typing	Type I'm / you're / she's when contraction is the objective.	target contraction	full form meaning-correct/form-needed; retry/reveal	Writing/spelling + Grammar/grammar_form
8	Listening detail	Hear three intros and match profile.	profile match	slow available; transcript after attempt	Listening/ listening_recognition
9	Micro-note	Read that contractions are common but full forms remain valid in many contexts.	No answer	note	usage
10	Dialogue completion	Complete Hi, I'm Maya. / Hello, ___ Daniel.	I'm	word builder or typing by support	dialogue audio; Writing/production + Grammar/grammar_form
11	Read aloud	Read I'm a student or sample-role sentence; mic pulses while recording.	model phrase	usable -> correct/try again; unavailable -> continue	Pronunciation/ intelligibility; no unrelated mastery
12	Lesson result	Resolve scored errors; view lesson outcome.	0 unresolved scored items	speech may be unassessed, not failed; relevant indicators only	progress + review seed

8. English Lesson 3 - Where are you from?

Objective: ask/name origin, form basic be questions and short answers, and use spelling support.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Visual lexical intro	View/listen to selected country cards.	the UK, Ukraine, Czechia, Poland, Germany, Spain	exposure	validated SVG flags + neutral scenes; audio
2	Chunk intro	Listen/read Where are you from? / I'm from Ukraine.	exposure	completed	2 en-GB voices
3	Question order	Transform You are from Poland into a question.	Are you from Poland?	correct reveals order note	Grammar/ grammar_form + use
4	Short answer	Answer Are you from...?.	Yes, I am. / No, I'm not.	objective-aware variants; punctuation tolerant	Grammar/ context_use
5	Profile choice	Read/hear She's from Spain and select profile.	correct profile	retry/reveal	character scenes; Vocabulary/meaning
6	Listening gist	Hear three introductions and identify country only.	configured country	no transcript first; accessibility override logged	Listening/ listening_gist
7	City and spelling support	Explore London, Prague, Kyiv, Warsaw and How do you spell it?.	listen/view	support exposure only	text cards + alphabet audio
8	Letter listening	Hear spelling of one short name/city and select string.	example M-A-Y-A	slow available	Listening/ listening_recognition + Writing/spelling
9	UK usage note	Notice that the UK normally takes the.	I'm from the UK.	note	en-GB/en-US shared usage metadata
10	Role-play prompt	Answer a system origin question using sample or consented profile.	I'm from [configured place].	typed route always; usable speech gets correct/try again; unavailable neutral	Speaking/ production if usable speech; Writing/production for typed fallback
11	Typing	Translate Звідки ви?.	Where are you from?	strict target; retry/reveal	Writing/production + Grammar/grammar_form
12	Lesson result	Resolve scored errors and review evidence.	0 unresolved scored items	partial speech evidence explicit; relevant progress only	progress + review seed

9. English Lesson 4 - Nice to meet you

Objective: combine greeting, name, role/origin and close in a short first-contact exchange and sample online profile.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Retrieval warm-up	Recall four prior chunks without full prompt.	adaptive targets	errors seed remediation	no new media; Vocabulary/Grammar
2	Dialogue listening	Hear a six-turn first meeting and answer gist.	Is this their first meeting?	answer then transcript toggle	2 voices + avatars; Listening/listening_gist
3	Dialogue chain: name	Complete greeting/name turn.	Hello. I'm...	support chosen by mastery	persistent dialogue state
4	Dialogue chain: origin	Complete origin question/answer.	Where are you from? / I'm from...	accepted variants	Vocabulary/ context_use + Grammar/grammar_form
5	Social-close chunk	Listen/repeat Nice to meet you / Nice to meet you too.	exposure	completed after exposure	audio

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
6	Pronunciation practice	Repeat Nice to meet you; mic pulses during recording.	sentence stress/intelligibility	usable -> correct/try again; unavailable -> retry/continue	Pronunciation/ pronunciation_target
7	Role-play	Respond to partner over three learner turns.	name + origin + social close	turn-level rubric; no hard speech gate	voice/avatar sync; Speaking if usable
8	Sample profile	Fill Name / Country / City for a fictional profile.	valid sample values	no PII required; typed route	Writing/context_use
9	Profile reading	Read a fictional profile and answer a short question.	Yes/No + configured detail	brief feedback	Vocabulary/ meaning + context_use; Reading assessment domain may aggregate separately in Level Test only.
10	Supported production	Create 3-4 short sentences from blocks or typing.	I'm... / I'm from... / I'm a...	rubric-based variants	Writing/production
11	Strategy note	Read: keep the exchange short and ask for repetition when needed.	No answer	note	learning strategy
12	Lesson result	Resolve scored errors and see module readiness.	0 unresolved scored items	speech status separate; show mistakes/completion and relevant indicators	progress + recap recommendation

10. English Module Recap - First day at a language club

New scenario: meet two new characters, understand their introductions, repair misunderstanding and complete a short exchange.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Mixed retrieval	Complete six due/core items in varied operations.	adaptive	no new instruction	review seed
2	Listening gist	Hear an arrival dialogue and identify who is new.	person choice	transcript after answer	Listening/ listening_gist
3	Listening detail	Identify name and country.	2 configured details	slow mode	Listening/ listening_detail
4	Greeting register	Classify morning/neutral/informal contexts.	Good morning / Hello / Hi	resolve all or reveal	original thumbnails; Vocabulary/register
5	Grammar correction	Correct I are....	I am / I'm	explain be form	Grammar/ grammar_form
6	Typing	Type Where are you from?.	exact target; punctuation tolerant	retry/reveal	Writing/production
7	Dialogue chain	Complete four turns including one repair phrase.	contextual responses	persistent prior turns	voices + avatars; item-specific Vocabulary/context_use + Grammar/grammar_form; Speaking only if a usable spoken turn is actually evaluated
8	Pronunciation pair	Practise one greeting and Nice to meet you; mic pulses on record.	model phrases	usable correct/try again; unavailable neutral; never unresolved blocker	Pronunciation/ pronunciation_target if usable

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
9	Final role-play	Complete four learner turns; no timer pressure.	greeting + name + origin + close	turn rubric; typed fallback gives Production, not Speaking	Speaking/ production if usable speech; Writing/production for typed fallback
10	Result	Fix scored mistakes and review module outcome.	N unresolved scored items	correct/reveal remediation; speech separate; broader skill summary MAY be shown	progress + immutable analytics

11. English review seed and media manifest

11.1. Review evidence

Observed result	Evidence update	Review consequence
Correct first try, no hint	positive evidence in target facet; contributes to global skill projection	normal scheduler expansion
Correct after retry	lower-quality positive facet evidence	earlier than clean retrieval
Reveal used	no independent mastery; reveal dependency recorded	high-priority near-term review
Listening wrong, typing right	Listening/listening_recognition practice-needed; Writing/spelling evidence unchanged	review in listening format
Usable speech matches target feature	Positive evidence to the Pronunciation facet declared by the activity evidence_map; Speaking only if the activity separately measures a usable spoken turn.	later contextual reuse
Usable speech mismatch	Practice-needed evidence only to the Pronunciation facet declared by evidence_map; no automatic penalty to meaning/grammar.	earlier listen-repeat/read-aloud
Low confidence/provider failure	no Speaking or Pronunciation mastery update	optional retry; no forced immediate queue
Role-play completed with usable audio	Speaking/interaction_goal evidence; Pronunciation separate	later contextual reuse
Typed role-play fallback	Writing/production evidence only	Speaking remains insufficient evidence
AI extra practice, free-form	practice telemetry only unless approved deterministic evaluator exists	may recommend review; no direct mastery mutation

11.2. English media minimum

Asset ID / pattern	Class	Generation specification	Use
IMG_ENG_M01_CLUB_01	generated scene	Original adult international club arrival; no visible text; alt text	L1 context
IMG_ENG_M01_MORNING_01	generated scene	Clear morning light; no text answer leakage	greeting image choice
IMG_ENG_M01_EVENING_01	generated scene	Evening entrance; no signage text	greeting image choice
AV_ENG_MAYA_01 / DANIEL_01 / TEACHER_01	3 avatars	Original recurring adults; character sheet; voice binding	dialogues and profiles
VEC_FLAG_GB / UA / CZ / PL / DE / ES	6 concrete SVGs	Validated flag geometry; each ID/version concrete after build	country cards
AUD_ENG_VOICE_F01 / M01 / F02	3 voice profiles	en-GB adult profiles; role, consent/licence and quality metadata	model and prompt audio
AUD_ENG_M01_WORDS_[concrete]	word/chunk audio	script checksum, en-GB voice, normal/slow policy, transcript	lexical items

Asset ID / pattern	Class	Generation specification	Use
AUD_ENG_M01_DIALOGUES_[concrete]	multi-speaker audio	turn assets + full mix; speaker/avatar/voice synchronization	L1/L4/Recap

11.3. English optional AI extension points

Trigger	Capability	Grounding / fallback
User taps Explain differently on a/an or contraction note	Grounded micro-explanation in Ukrainian + 1-2 A1 examples	Refs to ENG_K_A_AN / contraction knowledge; fallback to authored note.
Repeated error on be form	Adaptive hint or one extra controlled item	Hint policy + exact knowledge/facet; fallback to authored scaffold.
After L4/Recap	Bounded first-contact AI role-play, 3-5 turns	Allowed lexicon/functions from M01; scripted role-play fallback.
Short typed profile/self-intro	Advisory writing feedback	Focus on one/two highest-value A1 issues; no direct mastery without rubric.
AI unavailable/free quota exhausted	No AI call	Core lesson/review/recap unchanged.

ЧАСТИНА C. ČEŠTINA REFERENCE MODULE

12. Čeština reference module C-A1-M01

Field	Value
Title / scenario	Dobrý den! / První den v kurzu češtiny
Can-do	Greet formally/informally, introduce yourself, ask/give name and origin, use repair phrases and close a short exchange.
Variant	cs-CZ standard; ty/vy register explicit.
Grammar	být: jsem/jsi/jste/je; pronoun omission awareness; nominative/gender in identity nouns; origin chunks with z/ze + fixed forms.
Lexis	greetings, names, selected countries/languages, roles, repair phrases; numbers 0-20 as support pack.
Pronunciation	first-syllable stress without false vowel lengthening; vowel length, diacritics, č/š/ž/ě and ř awareness; slow audio.
AI extensions	optional grounded explanation of ty/vy, diacritics, origin forms; extra practice and bounded first-contact role-play.
Units	C-A1-M01-L01..L04 + C-A1-M01-R01 Module Recap.

12.1. Czech knowledge inventory

stable_code / group	Type	Canonical form	Scope	Function / evidence
CZE_K_DOBRY_DEN	chunk	Dobrý den.	active	formal/neutral greeting
CZE_K_AHOJ	chunk	Ahoj.	active	informal greeting
CZE_K_NASHLED	chunk	Na shledanou.	active	leave-taking
CZE_K_DEKUJI / PROSIM / PROMINTE	3 chunks	Děkuji. / Prosím. / Promiňte.	active	politeness and repair
CZE_K_NEROZUMIM	chunk	Nerozumím.	active	repair strategy
CZE_K_REPEAT	chunk	Můžete to zopakovat, prosím?	active	asking for repetition
CZE_K_NAME_Q_FORM	chunk	Jak se jmenujete?	active	formal personal information
CZE_K_NAME_Q_INF	chunk	Jak se jmenuješ?	receptive-active	informal personal information
CZE_K_NAME_A	chunk	Jmenuji se...	active	name answer
CZE_K_TESI	variant set	Těší mě. / Mě také. / Taky mě těší.	active	social close; last form informal
CZE_K_FROM_Q_FORM / INF	2 chunks	Odkud jste? / Odkud jsi?	active / receptive-active	formal/informal origin
CZE_K_FROM_A	pattern	Jsem z/ze [place form].	active	origin answer
CZE_K_BYT_JSEM / JSI / JSTE / JE	4 grammar	jsem / jsi / jste / je	active	Grammar/ grammar_form
CZE_K_REGISTER_TY_VY	usage	ty / vy and verb agreement	active	register + grammar
CZE_K_PRONOUN_DROP	usage	Já/Vy/Ty often omitted when clear	awareness	naturalness/context use
CZE_K_ROLE_STUDENT / TEACHER	2 lexical sets	student/studentka; učitel/učitelka	controlled	gender-linked identity
CZE_K_COUNTRY_CZ / UA / PL / DE / SK / ES	6 lexical/form sets	Česko/z Česka; Ukrajina/z Ukrajiny; Polsko/z Polska; Německo/z Německa; Slovensko/ze Slovenska; Španělsko/ze Španělska	active/support	origin forms; atomic IDs
CZE_K_LANG_CS / EN / UK / PL / DE / ES	6 lexical	česky / anglicky / ukrajinsky / polsky / německy / španělsky	receptive-active	language values
CZE_K_CITY_PRA / BRN / KYI	3 lexical/form sets	Praha/z Prahy; Brno/z Brna; Kyjev/z Kyjeva	support	city origin chunks

stable_code / group	Type	Canonical form	Scope	Function / evidence
CZE_K_NUM_00..20	21 lexical	nula..dvacet	support	recognition/listening only in M01; atomic IDs
CZE_K_PRON_STRESS	pronunciation	first-syllable stress	awareness	do not lengthen short stressed vowel
CZE_K_PRON_LENGTH	pronunciation	short/long vowel contrast	awareness	perception + pronunciation_target practice
CZE_K_PRON_DIACRITICS	pron/orthography	ě/š/č/ž/ý and encountered marks	active awareness	sound-spelling links
CZE_K_PRON_R	pronunciation	ř awareness	receptive	no production gate in M01
CZE_K_ORTH_NFC	orthography	canonical Unicode and Czech diacritics	active	strict in form tasks; no diacritic stripping

13. Čeština Lekce 1 - Dobrý den!

Objective: distinguish formal/informal greetings, use basic polite/repair phrases and notice first-syllable stress.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Context / scene	View first-day scene and hear Dobrý den / Ahoj.	No answer	intro -> completed	IMG_CZE_M01_CLASS_01 v1; 2 voices
2	Vocabulary intro	Listen to Dobrý den, Ahoj, Na shledanou, Děkuji, Prosím, Promiňte.	tap/listen	exposure	6 audio assets
3	Grammar/register classification	Sort Dobrý den / Ahoj by formal/informal context.	formal / informal	resolve all; note previews ty/vy	original thumbnails; Grammar/register
4	Listening match	Hear greeting and choose card.	Dobrý den / Ahoj / Na shledanou	retry/reveal	normal + slow; Listening/listening_recognition
5	Stress discovery	Tap first syllable in Dobrý, Ahoj, Promiňte; keep vowel length distinct.	syllable 1	correct reveals concise note	audio + syllable UI; Listening/listening_recognition (stress-position perception); no Pronunciation evidence
6	Listen and repeat	Repeat Dobrý den / Ahoj; mic pulses during recording.	model phrase; stress/intelligibility	usable -> correct/try again; unavailable -> retry/continue	Pronunciation/ pronunciation_target only if usable
7	Survival intro	Meet Nerozumím / Můžete to zopakovat, prosím?.	listen/read	exposure	2 audio assets
8	Phrase choice	Choose repair phrase when teacher speaks too quickly.	Můžete to zopakovat, prosím?	wrong explains function; retry/reveal	dialogue audio; Vocabulary/context_use (selection tests repair-phrase understanding; no Speaking evidence)
9	Word builder	Build DĚKUJI / PROSÍM with Czech keyboard support.	děkuji / prosím	Unicode NFC; diacritics strict for objective	Writing/spelling
10	Micro-note	Read that diacritics are part of spelling and stress does not equal vowel length.	No answer	note	text + audio replay

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
11	Dialogue chain	Complete formal teacher/learner greeting, thanks and leave-taking.	context choices	persistent turns; correct/reveal	avatars + voices; item-specific Vocabulary/context_use + Grammar/register; Speaking only if usable speech is evaluated
12	Lesson result	Resolve scored errors; review outcome.	0 unresolved scored items	speech may be unassessed, not failed; relevant indicators only	progress + review seed

14. Čeština Lekce 2 - Jsem / jste / jsi

Objective: use frequent být forms, distinguish vy/ty register and notice natural pronoun omission.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Character context	Hear Jsem Klára. / Vy jste Oksana? / Ty jsi Adam. / On je student.	exposure	completed	4 original avatars + audio
2	Interactive grammar	Complete Já ___ / Vy ___ / Ty ___ / On/Ona ___.	jsem / jste / jsi / je	staged explanation after response	Grammar/ grammar_form
3	Grammar/register mapping	Map teacher/new adult to vy and friend/peer to ty.	vy / ty	resolve all	original thumbnails; Grammar/register
4	Pronoun discovery	Compare Já jsem Klára and Jsem Klára; choose neutral intro.	Jsem Klára.	note: pronoun often omitted	Grammar/ context_use
5	Gender lexical cards	Match student/studentka and učitel/učitelka to profiles.	person-to-word	no full paradigm	original portraits; Vocabulary/meaning
6	Gap fill	Complete Ona ___ studentka / Vy ___ učitel?.	je / jste	retry/reveal	Grammar/ grammar_form
7	Typing	Type Jsem studentka or Jsem student for a fictional profile.	profile-compatible target	exact být form central	Writing/production
8	Listening detail	Hear three introductions and match name + role.	profile cards	slow available; transcript after attempt	Listening/ listening_recognition
9	Pronunciation practice	Repeat Jsem Klára / Vy jste Adam?; mic pulses while recording.	cluster/rhythm focus	usable correct/try again; unavailable neutral	Pronunciation/ intelligibility
10	Dialogue completion	Complete Dobrý den. ___ Petra.	Jsem	word builder/typing by support	Writing/production + Grammar/grammar_form
11	Micro-note	Read that jste is plural you and polite singular you.	No answer	note	Grammar/register awareness (no evidence by viewing)
12	Lesson result	Resolve scored errors and see outcome.	0 unresolved scored items	speech status separate; relevant progress only	progress + review seed

15. Čeština Lekce 3 - Odkud jste?

Objective: ask/give origin through frequent country/city chunks without requiring a full genitive paradigm.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Visual lexical intro	View/listen to six selected country names.	Česko, Ukrajina, Polsko, Německo, Slovensko, Španělsko	exposure	validated SVG flags + audio
2	Chunk intro	Listen/read Odkud jste? / Jsem z Ukrajiny.	exposure	completed	2 cs-CZ voices
3	Grammar/register pair	Classify Odkud jste? / Odkud jsi?.	formal / informal	context feedback	Grammar/register
4	Origin-pattern discovery	Match base country to z/ze origin form.	configured chunk	fixed forms now; systematisation later	Grammar/ morphology
5	Profile choice	Read/hear Je z Polska and choose profile.	correct profile	retry/reveal	character cards; Vocabulary
6	Listening gist	Hear three intros and identify country.	country choice	no transcript first; accessibility override recorded	Listening/ listening_gist
7	City support	Match Praha/z Prahy, Brno/z Brna, Kyjev/z Kyjeva.	base-to-origin form	guided; no full table	programmatic city cards
8	Diacritics awareness	Choose Česko vs Cesko and těší vs tesi.	correct Czech spelling	meaning may be understood; form strict	Writing/spelling facet
9	Language chunks	Listen/match Mluvím česky/anglicky/ukrajinsky.	configured language	controlled	audio; Vocabulary
10	Numbers support	Hear selected 0-20 items and match number card.	configured number	receptive/support only in M01	audio + programmatic cards; Listening/listening_recognition
11	Role-play + typing	Answer Odkud jste? and type Odkud jste?.	Jsem z/ze [accepted form]; Odkud jste?	typed always; usable speech correct/try again; unavailable neutral; typing retry/reveal	Speaking/ production if usable speech; Writing/production + Grammar/grammar_form for typed fallback
12	Lesson result	Resolve scored errors and review evidence.	0 unresolved scored items	partial speech evidence explicit	progress + review seed

16. Čeština Lekce 4 - Těší mě

Objective: combine greeting, name, register, origin and social close in formal and informal branches.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Retrieval warm-up	Recall six core chunks from L1-L3 in mixed modes.	adaptive	errors seed remediation	no new media
2	Dialogue listening	Hear a formal six-turn exchange and identify teacher/friend context.	context choice	answer then transcript	2 voices; Listening/listening_gist
3	Social-close chunks	Listen/repeat Těší mě / Mě také; Taky mě těší marked informal.	register-aware variants	exposure	audio

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
4	Formal dialogue chain	Complete Dobrý den -> Jak se jmenujete? -> Odkud jste? -> Těší mě.	configured turns	persistent turns	avatars + voices; item-specific Vocabulary/context_use + Grammar/register; Speaking only if usable speech is evaluated
5	Informal dialogue chain	Complete Ahoj -> Jak se jmenuješ? -> Odkud jsi?.	informal variants	context feedback	avatars; Grammar/register
6	Pronunciation practice	Repeat Těší mě with slow audio; notice ě/š; mic pulses during recording.	model phrase; intelligibility	usable -> correct/try again; unavailable -> retry/continue	Pronunciation/ pronunciation_target
7	Typing / diacritics	Type Těší mě / Děkuji.	strict Czech form	missing diacritics = form error; retry/reveal	Writing/spelling
8	Role-play branch	Choose formal/informal fictional context and respond over four turns.	register-consistent forms	semantic + register rubric; speech non-blocking	Speaking/ context_use + Speaking/register if usable speech; typed fallback -> Writing/context_use + Grammar/register
9	Sample profile	Fill fictional name/country/language fields with Czech labels.	valid sample text	no PII required	Writing/context_use
10	Profile reading	Read fictional profile and answer Je z Česka?.	Ano / Ne	brief feedback	Vocabulary/ meaning + context_use; Reading assessment domain may aggregate separately in Level Test only.
11	Strategy note	Read that endings expand gradually through high-frequency patterns.	No answer	note	metacognition
12	Lesson result	Resolve scored errors and see module readiness.	0 unresolved scored items	speech status separate; mistakes + relevant indicators	progress + recap recommendation

17. Čeština Module Recap - První den v kurzu

New scenario: greet teacher formally, meet a peer informally, give identity/origin and use a repair phrase when speech is too fast.

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
1	Mixed retrieval	Complete six core/due items.	adaptive	no new instruction	review seed
2	Grammar/register choice	Choose Dobrý den or Ahoj for teacher/peer scenes.	formal / informal	resolve all or reveal	original scenes; register
3	Listening gist	Identify teacher by dialogue/register.	character	transcript after answer	2 Czech voices; Listening
4	Listening detail	Identify name and country.	2 details	slow mode	Listening/ listening_detail
5	Grammar correction	Correct Vy jsi....	Vy jste...	explain register + verb form	Grammar/register
6	Origin pattern	Choose z Ukrajiny / z Česka / z Polska in context.	configured answer	no paradigm table	Grammar/ morphology

#	Activity / type	Learner action	Target / answer	Feedback and state	Evidence / media
7	Typing	Type Jak se jmenujete?.	strict diacritics where present; punctuation tolerant	retry/reveal	Writing/production
8	Pronunciation pair	Practise Dobrý den + Těší mě; mic pulses on record.	model phrases	usable correct/try again; unavailable neutral; never blocker	Pronunciation/ pronunciation_target if usable
9	Final role-play	Formal intro then brief informal peer exchange.	register + identity + origin + close	turn rubric; typed fallback not Speaking	Speaking/ context_use + Speaking/register if usable speech; typed fallback -> Writing/context_use + Grammar/register
10	Result	Fix scored mistakes and review module outcome.	N unresolved scored items	correct/reveal remediation; speech separate; broader skill summary MAY be shown	progress + immutable analytics

18. Czech review seed and media manifest

18.1. Czech-specific evidence

Case	Evaluation	Evidence consequence
Missing diacritics in controlled meaning-choice	Intended option can be identified because UI controls the choice.	Meaning evidence possible; no spelling evidence.
Missing diacritics in form-focused typing	Incorrect form; show exact correction; retry/reveal.	Writing/spelling practice-needed; Vocabulary meaning unchanged.
Já jsem vs Jsem in neutral intro	Both semantically accepted; shorter form may get naturalness note.	Meaning full; context_use nuance only.
ty/vy mismatch	Content may be understood, but register/form objective is incorrect.	Grammar/register error; no Vocabulary penalty.
Wrong country form after z/ze	Morphology error, not country-meaning error.	Grammar/morphology practice-needed.
Usable speech target-feature match	Positive quality-tagged pronunciation evidence; native accent imitation not required.	Pronunciation target facet may gain evidence; unrelated facets unchanged.
Usable speech mismatch	Practice-needed evidence only to the Pronunciation facet declared by evidence_map; no automatic penalty to meaning/grammar.	Pronunciation practice-needed only; no unrelated mastery penalty.
Low confidence/noisy audio/provider failure	No language judgment; speech_unavailable.	No mastery change; optional retry or continue.
AI explains ty/vy or diacritics	Advisory grounded explanation only.	No mastery by viewing; assistance telemetry only.

18.2. Czech media minimum

Asset ID / pattern	Class	Generation specification	Use
IMG_CZE_M01_CLASS_01	generated scene	Original modern adult Czech-course arrival; no visible text; alt text	L1 context
IMG_CZE_M01_FORMAL_01 / INFORMAL_01	2 generated scenes	Teacher/reception vs peer greeting; no stereotyped decoration	register classification
AV_CZE_KLARA_01 / ADAM_01 / OKSANA_01	3 avatars	Original recurring adults; character sheet; register role and voice binding	dialogues/profiles
VEC_FLAG_CZ / UA / PL / DE / SK / ES	6 concrete SVGs	Validated flag geometry; each ID/version concrete after build	country cards

Asset ID / pattern	Class	Generation specification	Use
AUD_CZE_VOICE_F01 / M01 / F02	3 voice profiles	cs-CZ adult profiles; role, licence/consent and quality metadata	model/prompt audio
AUD_CZE_M01_WORDS_[concrete]	word/chunk audio	script checksum, cs-CZ voice, vowel-length QA, transcript	lexical items
AUD_CZE_M01_DIALOGUES_[concrete]	multi-speaker audio	turn assets + full mix; speaker/avatar/voice synchronization	L1/L4/Recap

18.3. Czech optional AI extension points

Trigger	Capability	Grounding / fallback
User asks why jste, not jsi	Grounded ty/vy + verb-agreement explanation in Ukrainian	Refs to CZE_K_REGISTER_TY_VY + být forms; authored note fallback.
Repeated missing diacritics	Adaptive hint or extra diacritic-focused item	Exact orthography facet; no blanket ASCII correction; authored scaffold fallback.
Origin form confusion	Explain z/ze + memorized place form without teaching full paradigm	Grounded country/form set; no unsupported full-case expansion.
After L4/Recap	Bounded formal/informal first-contact role-play	M01 lexicon/register only; scripted branches fallback.
Short Czech self-introduction	Advisory writing feedback	Separate meaning, morphology, diacritics and register; no direct mastery without rubric.
AI unavailable/free quota exhausted	No AI call	Core Czech lesson/review/recap unchanged.

ЧАСТИНА D. SHARED RUNTIME, MEDIA AND QUALITY

19. Shared state logic and answer evaluation

19.1. State contract

State	Entry	Allowed actions	Exit / evidence rule
ready	activity rendered	answer, audio, slow audio, begin mic request	answering or permission_request
answering	learner interacts	submit, clear, play audio	checking
checking	scored answer submitted	cancel only on technical timeout	correct or incorrect
incorrect	objective evaluation failed	retry; reveal if policy allows	answering or revealed; no success evidence
correct	objective evaluation passed	next or allowed auto-advance	completed with positive evidence
revealed	system showed answer	next	completed; reveal dependency; no independent mastery
permission_request	learner starts optional speech	allow or deny	recording or speech_unavailable
recording	mic active; control pulses unless reduced motion	stop / endpoint / cancel	processing or speech_unavailable
processing	audio sent to configured evaluator	cancel after bounded timeout	speech_correct / speech_incorrect / speech_unavailable
speech_correct	usable result passes configured learner-facing threshold	next; optional repeat	completed; quality-tagged pronunciation/speaking evidence as mapped
speech_incorrect	usable, sufficiently confident mismatch	retry or continue; Auto-advance only after feedback dwell when enabled	completed when continued/auto-advanced; bounded negative/practice-needed evidence only to the exact Pronunciation facet declared by evidence_map (pronunciation_target or intelligibility); never success
speech_unavailable	denied, timeout, low confidence or provider failure	retry or continue	completed; no Speaking/Pronunciation mastery update
completed	allowed completion contract met	next activity	ready(next); event retains reason and versions

Auto-advance invariant
For scored items Auto-advance is allowed only after correct or revealed. For optional speech it is allowed only after an allowed speech completion state. It never turns a timeout, low-confidence result or skipped scored answer into success.

19.2. Answer normalization matrix

Dimension	English	Czech	Invariant
Unicode	NFC before comparison	NFC before comparison	canonical equivalence only; never repair language errors
Case	locale-aware casefold except proper-name objective	locale-aware casefold except proper-name objective	original answer retained in attempt
Whitespace	trim and collapse repeated spaces	trim and collapse repeated spaces	no word-order change
Punctuation	usually tolerant; objective can require	usually tolerant; objective can require	policy stored with evaluator version
Apostrophes	straight/curly normalized; contraction task may require apostrophe	not primary	no blanket character deletion
Full vs contraction	accepted by objective-specific variant set	n/a	meaning and form evidence separated

Dimension	English	Czech	Invariant
Diacritics	n/a	strict in form tasks; never stripped globally	meaning recognition may remain separate
Grammar/register	usage notes where relevant	formal/informal variant sets	ty/vy mismatch stays visible
Morphology	be/article feature evaluation	inflected place and být forms	do not normalize wrong form into correct
Speech	language + selected variety + phrase version	cs-CZ + accepted phrase/form	usable result may produce correct/try again; confidence/provider failure stored as unavailable, not truth

19.3. Pronunciation interaction contract

Pronunciation is a learner-facing interaction, but the speech evaluator is not treated as infallible. The UX may provide a simple correct/try again result only when the signal is usable and confidence is above the calibrated threshold. Technical uncertainty is never presented as a language error.

Phase	Default UX	Evidence rule
Idle	Microphone control available; no permission prompt yet.	No evidence.
Record	Explicit tap requests permission; microphone control pulses/animates; static/text state under reduced-motion.	No evidence until evaluation.
Usable + correct	Control/feedback shows Correct / Далі; Auto-advance only if enabled.	Positive pronunciation facet evidence; Speaking only if task also measures a produced turn.
Usable + mismatch	Show error/try-again cue with short target-specific hint; retry or continue per activity policy.	Practice-needed only in the exact Pronunciation facet declared by evidence_map (pronunciation_target or intelligibility); no penalty to meaning/grammar/vocabulary.
Low confidence / silence / denial / outage	speech_unavailable; offer model audio, retry and continue.	No Speaking/Pronunciation mastery update.
Replay / slow	Normal and slow model audio remain available.	Playback alone is not mastery evidence.

20. Media generation and asset QA

Stage	Required behavior	Blocking validation
1. Requirement	Activity declares image/audio/avatar/vector need and accessibility role.	Missing requirement for semantically required asset.
2. Prompt/script	Deterministic content spec; original concept; no Babbel source material.	Copied/proprietary wording or missing script checksum.
3. Generation/licensing	Generate original scene/character; validated SVG for precision; licensed/consented voice where applicable.	Unknown rights/provenance.
4. Automated checks	Dimensions, MIME, checksum, duration/loudness, unwanted text, duplicate/link integrity.	Invalid file, duplicate, unresolved text or script mismatch.
5. Language/media QA	Semantic correctness, no answer leakage, cultural review, character consistency, audio/text match.	Incorrect language, misleading visual or speaker mismatch.
6. Accessibility QA	Alt text/decorative flag, transcript/caption, contrast and non-audio instruction path.	Required alternative missing.
7. Persist/version	Store asset_id, asset_version, class, locale/variety, prompt/script checksum, voice/character, rights/provenance, QA status.	Required metadata absent.
8. Bind/publish	Activity references concrete asset_id + version; filename/wildcard is not application logic.	Unresolved/non-approved asset or wildcard binding.

20.1. Required asset metadata

Field	Purpose	Rule
asset_id / asset_version	stable binding and rollback	both concrete and immutable after publication
class / locale / variety	correct renderer and voice policy	en-GB/en-US/cs-CZ explicit where relevant
prompt_or_script_checksum	reproducibility and audio/text match	changes create a new version
voice_id / character_id / turn_id	dialogue synchronization	required for role-bound audio
rights / provenance	copyright and consent trace	approved before publish
qa_status / reviewer / timestamp	publish gate evidence	status must be approved
alt_text / decorative / transcript	accessibility	semantic media must have an alternative
duration / speed / loudness	audio behavior and QA	normal/slow relation explicit
generation_provider/model/spec_version	generation trace and regeneration	internal metadata; provider URL is not lesson dependency

20.2. AI-generated media policy

- AI/image/TTS providers are replaceable adapters; provider URLs and SDK types never become canonical content identifiers.
- Generated media must pass the same originality, semantic, accessibility, linguistic and rights/provenance gates as any other asset.
- Bulk generation uses deterministic requirements, quotas, retry/cost telemetry and versioned prompt/style/voice specifications.
- If AI generation is unavailable, already-published lessons continue to use stored platform-controlled assets. Live generation is never a normal lesson dependency.
- Real-person voice/face cloning is not used without separate lawful basis and explicit consent.

21. Accessibility and privacy contract

Area	Required behavior	Failure condition
Keyboard/touch	All scored/navigation actions work by keyboard and touch with visible focus and adequate target size.	Mouse-only drag or unreachable control.
Audio alternatives	Instructions exist in text; transcripts/captions available. First-attempt hiding may be overridden by accessibility mode and logged.	Audio is sole carrier of required instruction.
Images	Semantic images have concise alt text; decorative images marked decorative without leaking answers.	Missing/misleading alternative.
Mic	Permission requested just in time; denial/unavailable/provider failure gives retry/model audio/continue.	Lesson/Recap blocked or learner marked failed.
Motion	Mic pulse and feedback respect reduced-motion; status also textual/static.	Animation is the only state signal.
Colour	Correct/error/register states use text/icon/shape in addition to colour.	Meaning conveyed only by colour.
Personal data	Fictional sample profile is default. Real name/country insertion optional and purpose-limited.	PII required to complete objective.
Recordings	Raw audio transient by default; derived metadata follows retention policy.	Undisclosed indefinite retention.
AI privacy	Learner text/audio sent to AI only for declared capability, minimized and provider-policy approved; sensitive data not required.	Hidden provider use, unnecessary PII, or retention/training terms outside approved policy.

22. Analytics and regression test suite

Every event includes language profile/user_course, content/item/evaluator/mastery/scheduler versions, operation, support level, attempt outcome, reveal/skip flags and error/facet category where applicable. Speech additionally stores permission/provider/confidence quality. AI events store capability/provider/model/prompt-template/grounding version and cost/quota class without turning model output into definitive learner truth.

Test	Scenario	Expected result
T01	English contraction target; user enters full form	Meaning accepted; form feedback requests contraction; retry/reveal; no dead end.

Test	Scenario	Expected result
T02	Czech strict typing omits diacritics	Form incorrect; exact correction; earlier spelling review; meaning not erased.
T03	Czech neutral intro uses Já jsem instead of Jsem	Accepted; optional naturalness note; no penalty.
T04	Formal Czech question answered with informal verb form	Grammar/register/Grammar error only; retry/reveal.
T05	Usable speech above threshold matches target feature	Learner sees correct/Далі; positive Pronunciation facet evidence; no unrelated mastery.
T06	Usable speech above threshold mismatches target feature	Learner sees try-again/error cue; retry/continue per policy; practice-needed only in target facet.
T07	Mic denied or absent	speech_unavailable; model audio/retry/continue; Speaking/Pronunciation insufficient evidence, not failed.
T08	Speech provider timeout or low confidence	No language judgment/mastery update; retry/continue; provider state logged.
T09	Reveal used	Item completes; no independent mastery; near-term review.
T10	Wrong scored answer with Auto-advance ON	No advance until correct or reveal.
T11	Reload during lesson	Returns to last unresolved activity; accepted attempts preserved.
T12	Required media missing/unapproved	Publish blocked; runtime never shows broken production placeholder.
T13	Asset version changed after publication	Old attempt retains old version; new content references explicit new version.
T14	Switch English -> Czech -> English	Independent progress/mastery/review restored exactly.
T15	Czech review candidate requested in English course	Candidate rejected by language isolation invariant.
T16	Slow audio	Same script at slower pitch-preserved rate or approved slow asset; transcript unchanged.
T17	Remediation	Only scored unresolved items repeat; speech never blocks.
T18	en-US selected	Voice/notes/accepted variants follow en-US while shared module structure remains.
T19	Unicode canonically equivalent Czech input	NFC normalization matches; original input retained.
T20	Czech wrong diacritic or wrong case form	Not normalized into correct; objective-specific facet error recorded.
T21	Keyboard-only and touch-only completion	All lesson/Recap paths reachable with visible focus/state.
T22	Transcript accessibility override before first listening attempt	Completable; support level recorded, not treated as cheating.
T23	Sample profile path	No PII requested; Writing objective completes with fictional data.
T24	Unexpanded ID range or media wildcard at publish	Build/publish rejected with precise manifest error.
T25	One lexical item: meaning correct, spelling wrong	Meaning facet improves; spelling facet does not; global Vocabulary projection remains derived.
T26	Lesson result after partial evidence	Shows mistakes/completion and relevant indicators; does not fabricate six numeric scores.
T27	AI disabled globally	Both English and Czech reference modules complete with identical core outcomes.

Test	Scenario	Expected result
T28	AI quota/provider unavailable during Explain differently	Authored explanation fallback; no dead end or mastery penalty.
T29	AI explanation request	Output is grounded in declared knowledge/content refs and stays within A1 scope; unsupported claim rejected/fallback.
T30	AI free-form extra practice answer	Practice telemetry recorded; mastery unchanged unless activity uses approved evaluator/rubric.
T31	AI admin/content generator produces draft	Draft cannot publish without normal validation/review/media gates; action audited.
T32	AI role-play unavailable mid-session	Session exits to scripted fallback or completion path; accepted core attempts remain durable.

23. Definition of Done and implementation hand-off

- Both reference modules load through one versioned content schema and shared Lesson Engine renderers.
- Every listed activity type has renderer, evaluator/completion contract and automated test.
- Progress survives reload/login and is isolated by language_profile/user_course and target language.
- English contractions/variety and Czech diacritics/register/morphology use language-aware evaluation, not engine forks.
- Two-level mastery persists relevant item facets and derives six global skill projections without conflating them.
- Normal/slow audio, transcript policy and multiple voice bindings work for all relevant items.
- Speech permission, pulsing recording state, usable correct/try-again feedback and neutral unavailable states are non-blocking and privacy-compliant.
- All required media has concrete asset_id + asset_version, approved QA, rights/provenance and accessibility metadata.
- Each Module Recap is a new integrated scenario with deterministic remediation; full six-skill display is optional, not mandatory.
- Optional AI learning/admin extensions use provider-neutral capability contracts, grounding, quotas/fallback and do not become core dependency.
- Reveal/retry/skip/Auto-advance rules have no dead ends and preserve immutable evidence.
- Mobile and desktop render the same content semantics with responsive composition and accessibility parity.
- The 32-scenario regression suite passes for English, Czech, two-level mastery, pronunciation and AI fallback/grounding.
- A defect list records every gap; contract change creates controlled revision of governing specifications before scaling.

23.1. Implementation order

Order	Work package	Exit evidence
1	Freeze current v1.6 content mappings, evaluator interfaces, two-level mastery evidence map and AI capability boundaries.	schema fixtures + validator + version contract
2	Build production-oriented project skeleton, persistence and immutable attempt/evidence/mastery path from Technical v1.7 / Design v1.7.	reload/login/language-isolation tests
3	Implement shared activity renderers, correct-or-reveal flow and pronunciation branch with pulsing/correct/try-again/unavailable states.	component/state tests including failures
4	Build media manifest tooling and produce approved English/Czech assets.	resolved concrete asset versions and QA records
5	Load E-A1-M01 and run vertical-slice QA.	English evidence report and defect list
6	Load C-A1-M01 without engine fork and run vertical-slice QA.	Czech evidence report and defect list
7	Add optional AI Gateway/capabilities behind feature flags and zero-paid-budget default; prove non-AI fallback.	AI contract/grounding/quota/fallback tests
8	Run responsive, accessibility, privacy and 32-scenario regression QA.	signed release-candidate report
9	Resolve gaps, freeze reference baseline, then author/implement M02-M12 in controlled batches.	approved baseline revision and production plan

Implementation start condition
After final audit and P0 owner lock of this v1.6, programming may begin. Approval does not mark the engine validated; validation is achieved only when implementation evidence and regression suite pass. No separate throwaway one-folder

prototype is a required gate.

24. Sources and change control

- Council of Europe (2020). Common European Framework of Reference for Languages: Learning, teaching, assessment - Companion Volume.
- Council of Europe. CEFR resources and the action-oriented approach.
- English Profile. English Grammar Profile and English Vocabulary Profile.
- NPI / Čeština pro cizince. Referenční popis češtiny pro účely zkoušky z českého jazyka pro trvalý pobyt v ČR - úrovně A1, A2; Soupis lexikálních jednotek A1.
- Internal baseline: Lingvessa Formal Specification v1.7; Lingvessa Technical Specification v1.7; Technical Design v1.7; Curriculum Blueprint A1 v1.6; AI Integration Specification v1.4; Implementation Plan v1.4.

Source use and originality
External sources support level descriptors and language-specific scope only. They are not sources for copying proprietary course text, activity sequences, media, branding or trade dress. This Reference Module Design is an original Lingvessa artifact.

Change type	Required control
Knowledge/lesson objective	curriculum + language lead approval; trace and migration impact
Accepted variant/evaluator	language + technical QA; versioned regression result
Mastery facet/evidence map	technical + pedagogical review; model/version compatibility and replay test
Speech behavior	non-blocking invariant, privacy/accessibility review, calibrated threshold and provider-failure tests
AI capability/prompt/grounding	AI Integration policy + pedagogical/privacy/technical review; feature flag and fallback test
Asset/script	new concrete version, QA, rights/provenance and accessibility refresh
State/completion contract	Technical Specification/Design revision if shared contract changes

Approval record
This v1.6 is the owner-locked reference baseline as of 10.09.2026; implementation P0 is authorized; future changes create a controlled version and do not overwrite historical evidence.

Controlled baseline synchronization - 09.09.2026

Normative baseline
This document is part of the synchronized Lingvessa owner-locked implementation baseline; P0 is authorized as of 10.09.2026. Earlier current-version references are superseded by the exact set below; historical "replaces" statements remain historical only.

Document	Version	Authority
Lingvessa_Formal_Specification	v1.7	Product scope, pedagogy, user-facing invariants
Lingvessa_Technical_Specification	v1.7	Normative technical requirements and interfaces
Lingvessa_Technical_Design	v1.7	Concrete implementation contracts, DDL/state/API design
Lingvessa_Curriculum_Blueprint_A1	v1.6	A1 pedagogical scope and module architecture
Lingvessa_Reference_Module_Design	v1.6	Executable E-A1-M01/C-A1-M01 fixtures
Lingvessa_AI_Integration_Specification	v1.4	AI-specific policy, data, routing, permissions
Lingvessa_Implementation_Plan	v1.4	Sole execution-order authority; no scope override

Decision	Topic	Fixed rule
BL-001	Baseline set	Formal 1.7 / Technical 1.7 / Design 1.7 / Curriculum A1 1.6 / Reference 1.6 / AI 1.4 / Plan 1.4
BL-002	Precedence	Domain-specific governing document controls; unresolved conflict -> owner decision + ADR/spec revision before code merge.
BL-003	Identity contract	id = UUID; stable_code = immutable human-readable semantic code; version_code = immutable revision label. Never put stable_code in *_id.

BL-004	Mastery contract	Exactly six global skills; canonical facet registry in this release. Reading is assessment-only domain, not seventh mastery skill.
BL-005	Language isolation	Composite DB FKs/checks/trigger guards enforce user + language_profile + language lineage for learner state and learner AI.
BL-006	Pronunciation states	speech_correct / speech_incorrect / speech_unavailable are canonical. Incorrect is non-blocking; Auto-advance occurs only after feedback dwell and never records success.
BL-007	Speech lifecycle	Asynchronous speech_job lifecycle with idempotent final attempt settlement and transient audio cleanup.
BL-008	AI API/data model	Canonical /api/v1/ai/explain|hint|practice|writing-feedback|conversations plus unified ai_* physical model.
BL-009	Monorepo	One /packages/provider-adapters package with ai/, speech/, tts/, image/, email/, storage/ sub-adapters; /api-client and /ui retained.
BL-010	Execution order	Implementation Plan v1.4 P0-P13 is the sole milestone numbering authority.
BL-011	Learner AI opt-in	Learner AI is OFF by default until explicit opt-in/consent. Effective enablement order is global -> capability -> provider/policy -> learner setting/consent -> quota.
BL-012	OpenAPI	Pin API description to OpenAPI Specification 3.1.2 for v1 baseline; upgrades require ADR/contract diff.

Precedence: Formal controls product/pedagogical scope; Curriculum controls A1 pedagogy; Technical Specification controls normative technical requirements; Technical Design controls concrete implementation detail; AI Specification controls AI-specific behavior; Reference Module Design is the executable fixture; Implementation Plan controls execution order only. Any unresolved conflict blocks implementation until an owner decision and controlled revision/ADR.

Canonical mastery/facet registry

Machine-code rule
Only canonical facet codes may appear in evidence_map. Every (skill, facet) pair MUST be in the eligible-pair matrix; invalid pairs are publish-blocking. Selection-only activities never emit Speaking evidence.

Facet code	Meaning
recognition	Recognition of a known form/choice
meaning	Meaning comprehension
recall_l2_to_l1	Target-language -> explanation-language recall
recall_l1_to_l2	Explanation-language -> target-language recall
production	Independent target-language production
spelling	Spelling of lexical/form surface
orthography	Language-specific orthographic correctness
listening_recognition	Recognition from audio
listening_gist	Gist from audio
listening_detail	Specific detail from audio
pronunciation_target	Named pronunciation feature
intelligibility	Understandability of learner speech
context_use	Appropriate use in context
collocation	Chunk/collocation integrity
grammar_form	Grammar form
word_order	Word order
morphology	Inflection/morphology
register	Grammar/register/formality
variant_awareness	en-GB/en-US or accepted language variant awareness
interaction_repair	Repair/clarification move
interaction_goal	Goal completion in interaction

Facet-to-global-skill aggregation

Global skill	Eligible facets (only when evidence_map explicitly targets the skill)
Vocabulary	recognition, meaning, recall_l2_to_l1, recall_l1_to_l2, production, spelling, context_use, collocation,

	variant_awareness, register
Grammar	grammar_form, morphology, word_order, register, context_use
Listening	listening_recognition, listening_gist, listening_detail, meaning
Speaking	production, context_use, register, interaction_repair, interaction_goal, intelligibility
Writing	production, spelling, orthography, grammar_form, context_use, register
Pronunciation	pronunciation_target, intelligibility

Reading is not a seventh global mastery skill. Reading tasks may emit item-facet evidence to Vocabulary and/or Grammar according to the objective. The optional Level Test may report a separate assessment_domain=reading subscore; that subscore is assessment-only and is not persisted as skill_mastery_states.

Canonical identity and version contract

Field	Type	Example	Rule
id	UUID	018f...	Opaque database/API identity.
stable_code	string	E-A1-M01-L01	Immutable human-readable semantic code; never stored in *_id.
version_code	string	1.4 / 2026-09-05.1	Immutable revision label for a versioned entity.
provider_external_id	string nullable	vendor-model-x	External/provider identifier only in adapter/config boundary.

API/JSON examples MUST use UUID-shaped values in *_id fields. Content manifests MAY use stable_code for authoring references, but the build/publish step resolves them to UUID + version_code and rejects unresolved codes.

Canonical pronunciation state and asynchronous speech-job contract

State	Meaning	Completion / evidence / Auto-advance
speech_correct	Usable, sufficiently confident target-feature match	Positive bounded Pronunciation evidence to the exact facet declared by evidence_map (pronunciation_target or intelligibility). Auto-advance allowed after feedback dwell if setting ON; otherwise show “Далі”.
speech_incorrect	Usable, sufficiently confident mismatch	Bounded negative/practice-needed evidence only to the exact Pronunciation facet declared by evidence_map (pronunciation_target or intelligibility). Non-blocking. Auto-advance may proceed after feedback dwell if setting ON; otherwise show Retry + Continue. Never counts as success.
speech_unavailable	Mic denied/absent, timeout, low confidence, provider failure	No mastery update. Non-blocking. Auto-advance may proceed after neutral status dwell if setting ON; otherwise Retry + Continue.

Canonical state names are speech_correct, speech_incorrect and speech_unavailable. Legacy labels speech_feedback_correct and speech_feedback_retry are invalid.

speech_job lifecycle

created -> uploaded -> queued -> processing -> succeeded | unavailable | failed | cancelled | expired. POST /api/v1/speech/jobs with Idempotency-Key + attempt_id creates the owner/profile/item-bound job or returns the existing job and a single-use transient upload target. PUT to the returned upload_url uploads encrypted transient audio; POST /api/v1/speech/jobs/{job_id}/submit transitions uploaded -> queued. GET /api/v1/speech/jobs/{job_id} returns owner-scoped status/result. POST /api/v1/speech/jobs/{job_id}/cancel is idempotent and settles the learner attempt exactly once as speech_unavailable(reason=user_cancelled); Continue is then the normal LessonPlayer Next action and creates no new speech attempt. succeeded settles exactly once as speech_correct or speech_incorrect using the activity evidence_map facet; unavailable/failed/expired settle exactly once as speech_unavailable after retry policy. Network retries reuse the same attempt_id/idempotency key/job; internal provider retries create no learner attempt. Learner Retry after any terminal learner-facing result creates a NEW attempt_id and speech_job. Signed internal provider callbacks are owner/job/version checked; late/duplicate callbacks after settlement are logged and ignored. Temporary raw audio is deleted immediately after terminal settlement/cancel or TTL (maximum 15 minutes by default).

Canonical AI API and persistence mapping

API	Purpose
POST /api/v1/ai/explain	Grounded explanation
POST /api/v1/ai/hint	Bounded hint
POST /api/v1/ai/practice	Extra practice
POST /api/v1/ai/writing-feedback	Writing feedback
POST /api/v1/ai/conversations	Create bounded conversation session

POST /api/v1/ai/conversations/{session_id}/messages	Conversation turn
DELETE /api/v1/ai/conversations/{session_id}	Delete/close learner AI session
Logical concept	Physical entity/table
AIProviderProfile	ai_provider_profiles
AIModelProfile	ai_model_profiles
AIPolicy / permissions / budget eligibility	ai_policies
AIPromptTemplate	ai_prompt_templates
AIRequestRecord / AIExecution / AIRun	ai_runs
AISession	ai_sessions
AIMessage	ai_messages
AIArtifact	ai_artifacts
AIUsage / budget consumption	ai_usage_buckets + audit_events
AIToolAction	ai_tool_actions

Learner AI is OFF by default. Effective enablement order: global AI flag -> capability flag -> approved provider/model/data policy -> learner opt-in and required consent -> quota/budget. Any failed check returns deterministic fallback; no paid fallback is implicit.

Canonical execution authority and milestone map

Milestone	Authority
P0	Governance freeze, exact baseline, decision log, full MUST trace matrix
P1	Monorepo/bootstrap/CI/config
P2	Data, UUID/stable_code/version_code, language isolation, publication
P3	Lesson Engine + attempts + deterministic state
P4	Two-level Mastery + Review + mistakes + collections
P5	Media/audio/speech + async speech jobs
P6	English E-A1-M01
P7	Czech C-A1-M01 + bilingual isolation proof
P8	Admin/CMS + publication + Admin AI Assist foundation
P9	Optional learner AI + AI Auto allowlist
P10	Onboarding/Home/Explore/Course/Progress/Profile/Settings/ notifications + persistence
P11	Reference hardening + 32 scenarios + technical gates; freeze reference baseline
P12	A1 M02-M12 controlled rollout + pilot/calibration + optional A1 Level Test
P13	Full-system hardening/launch gate: security, privacy, performance, restore/DR, release provenance, legal/product readiness

Only Lingvessa_Implementation_Plan v1.4 uses P0-P13 numbering. Any older P-number labels in other specifications are descriptive history and MUST be mapped to these milestones rather than treated as execution authority.

Canonical external references

- OpenAPI Specification 3.1.2
- CEFR Companion Volume
- Council of Europe - CEFR
- English Profile
- Čeština pro cizince

OpenAPI here means the OpenAPI Specification for describing HTTP APIs; it is unrelated to OpenAI as an AI provider. OpenAI remains only a possible future provider behind the provider-agnostic AI adapter and is not required by the v1 baseline.

Approval record

Поле	Значения
Baseline status	OWNER-LOCKED IMPLEMENTATION BASELINE / P0 AUTHORIZED
Approval date	10.09.2026
Approver	Product owner
Candidate version	v1.6
Accepted exceptions	None
Execution authority	Lingvessa_Implementation_Plan v1.4; P0 must be closed before P1.

Canonical machine-readable source: /packages/activity-schemas/mastery-registry.json. CI MUST validate every evidence_map against the registry and eligible skill-facet pair matrix; copied tables in specifications are checksum-checked mirrors.